

CPU scheduling in OPERATING SYSTEMS –2026

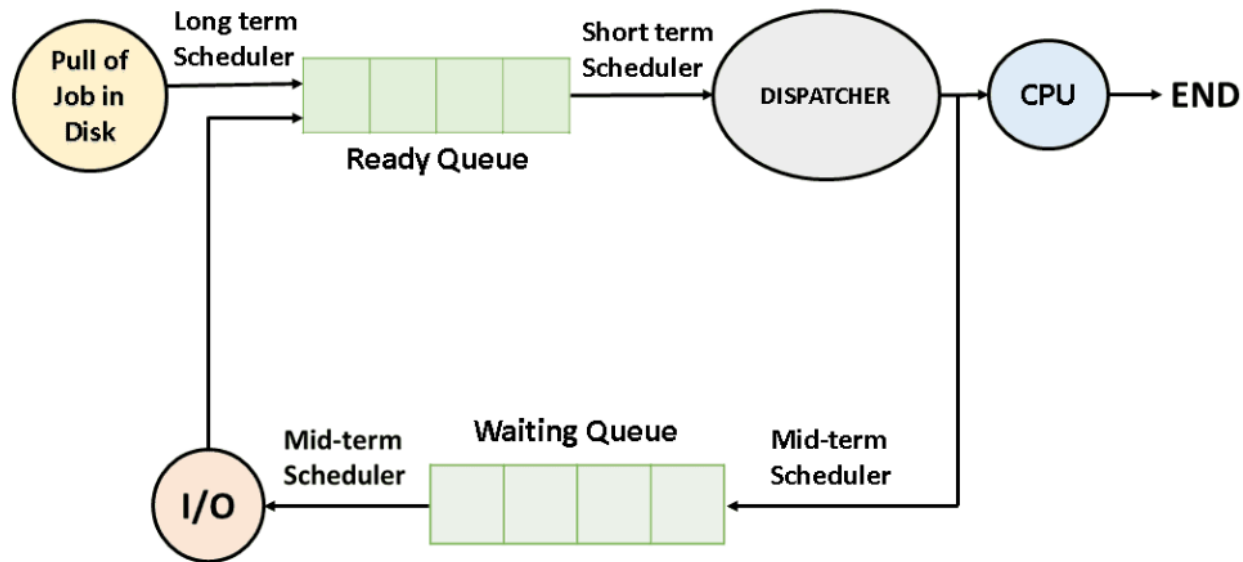
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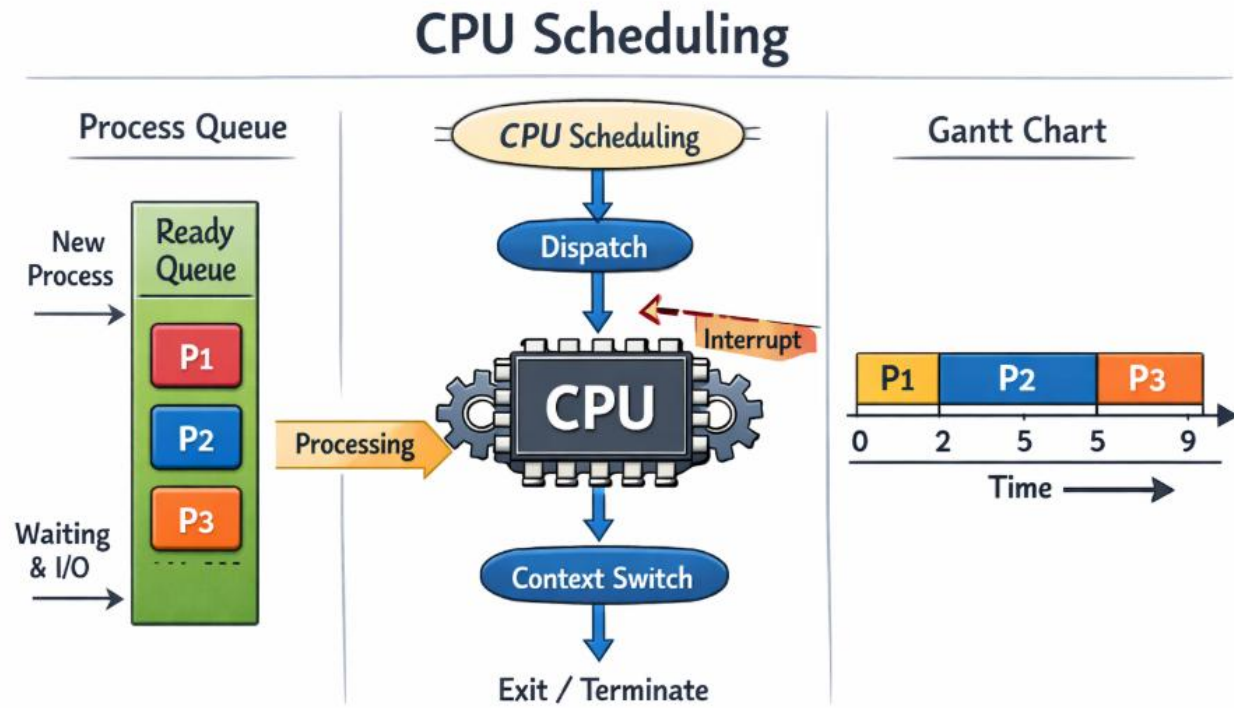
CPU scheduling

CPU Scheduling in Operating Systems



What is the CPU scheduling

- When many programs want the CPU at the same time, CPU scheduling chooses one to run first.



What is the CPU scheduling

Why CPU Scheduling is Needed?

- The CPU can run **only one process at a time.**
- There are **many processes** in memory.
- Scheduling helps to:
 - Use the CPU efficiently.
 - Reduce waiting time.
 - Make the system fast and responsive.

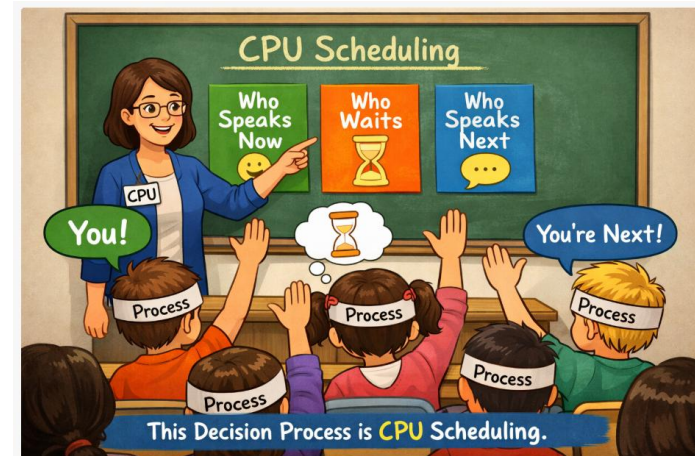
What is the CPU scheduling

Example (Real Life)

- Imagine **one teacher (CPU)** and **many students (processes)** lifting their hands.

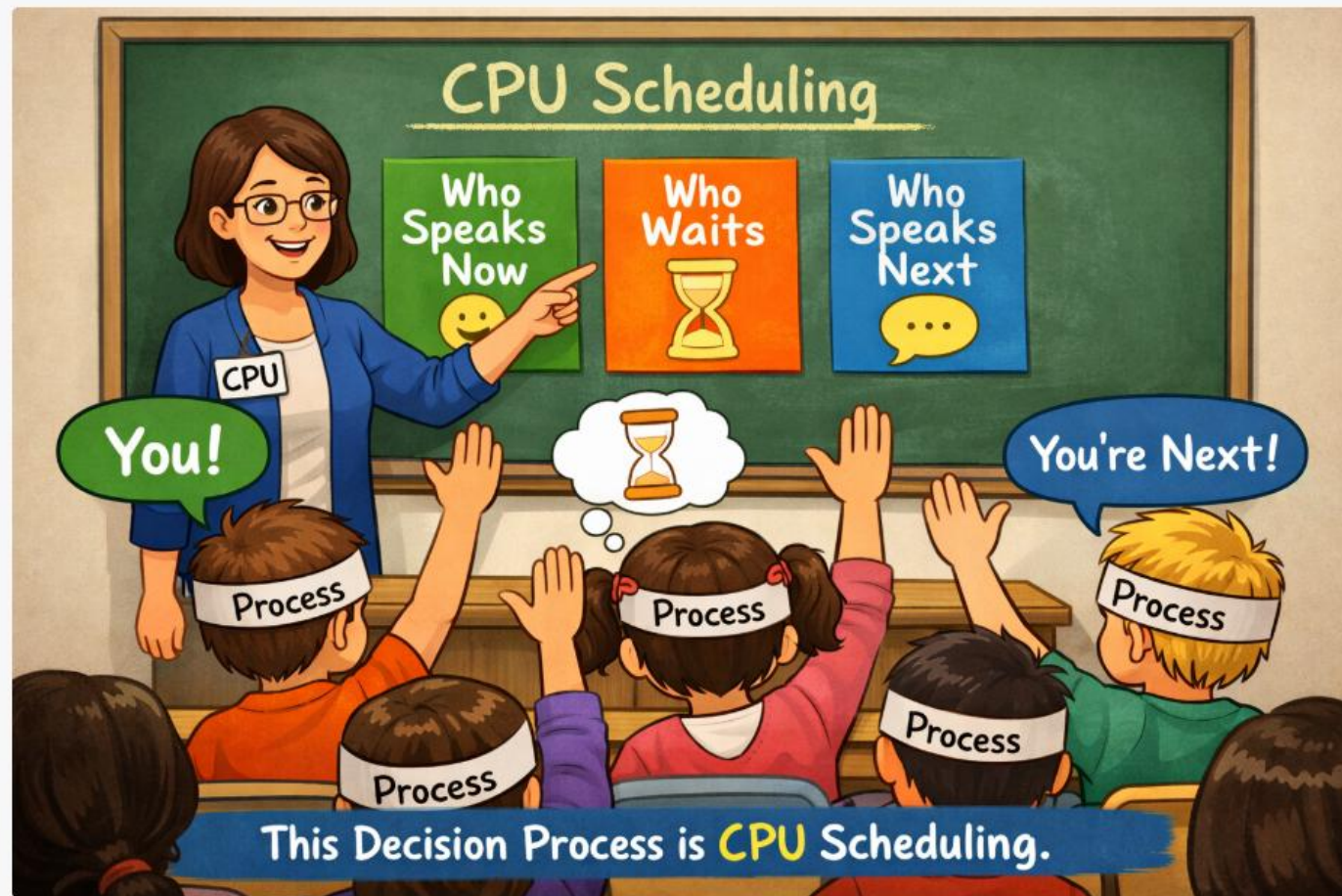
The teacher decides:

- Who speaks now.
- Who waits.
- Who speaks next.



- This decision process is **CPU scheduling**.

What is the CPU scheduling



Main Situations When CPU Scheduling Occurs

When a Process Finishes Execution.

- The running process completes its work.
- CPU becomes free.
- OS selects **another process** from the ready queue.
- Example: **A program ends → OS chooses the next program to run.**

Goal of CPU Scheduling

- Maximize CPU utilization.
 - Minimize waiting time.
 - Minimize turnaround time.
 - Improve system performance
- CPU scheduling is the process by which the OS **selects one process from the ready queue to occupy the CPU.**

Types of process schedulers

Three types of process schedulers

- Long term or Job Scheduler.
- Short term or CPU Scheduler .
- Medium-term Scheduler.

Types of process schedulers

1. Long-Term Scheduler (Job Scheduler)

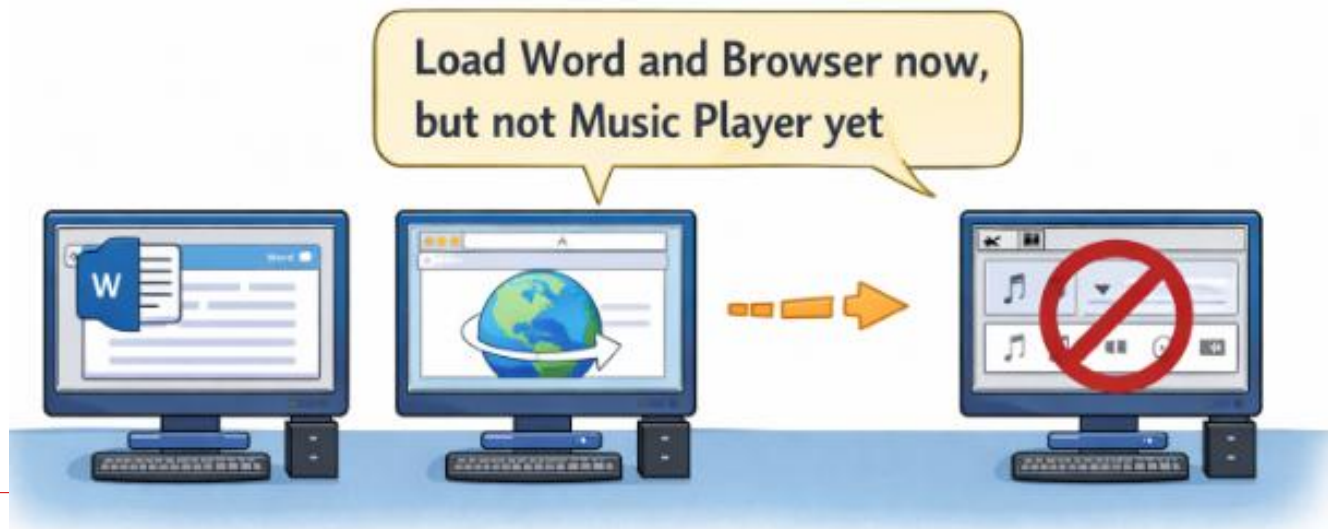
What it does:

- OS Decides **which jobs/processes are allowed to enter the system.**
- **Loads processes from disk into main memory (RAM).**

Types of process schedulers

Example:

- **In a computer lab, it decides:**
- “Load Word and Web Browser now, but not Music Player yet”.

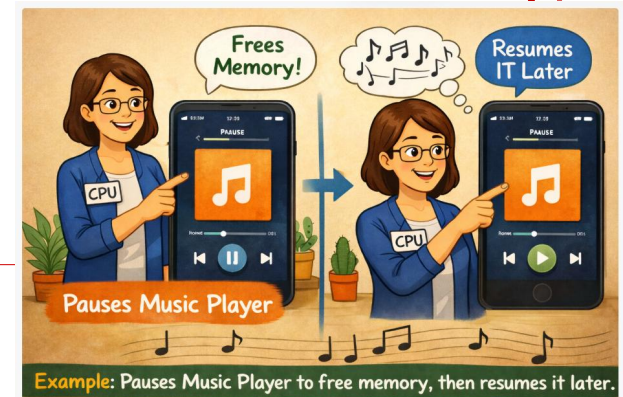


Types of process schedulers

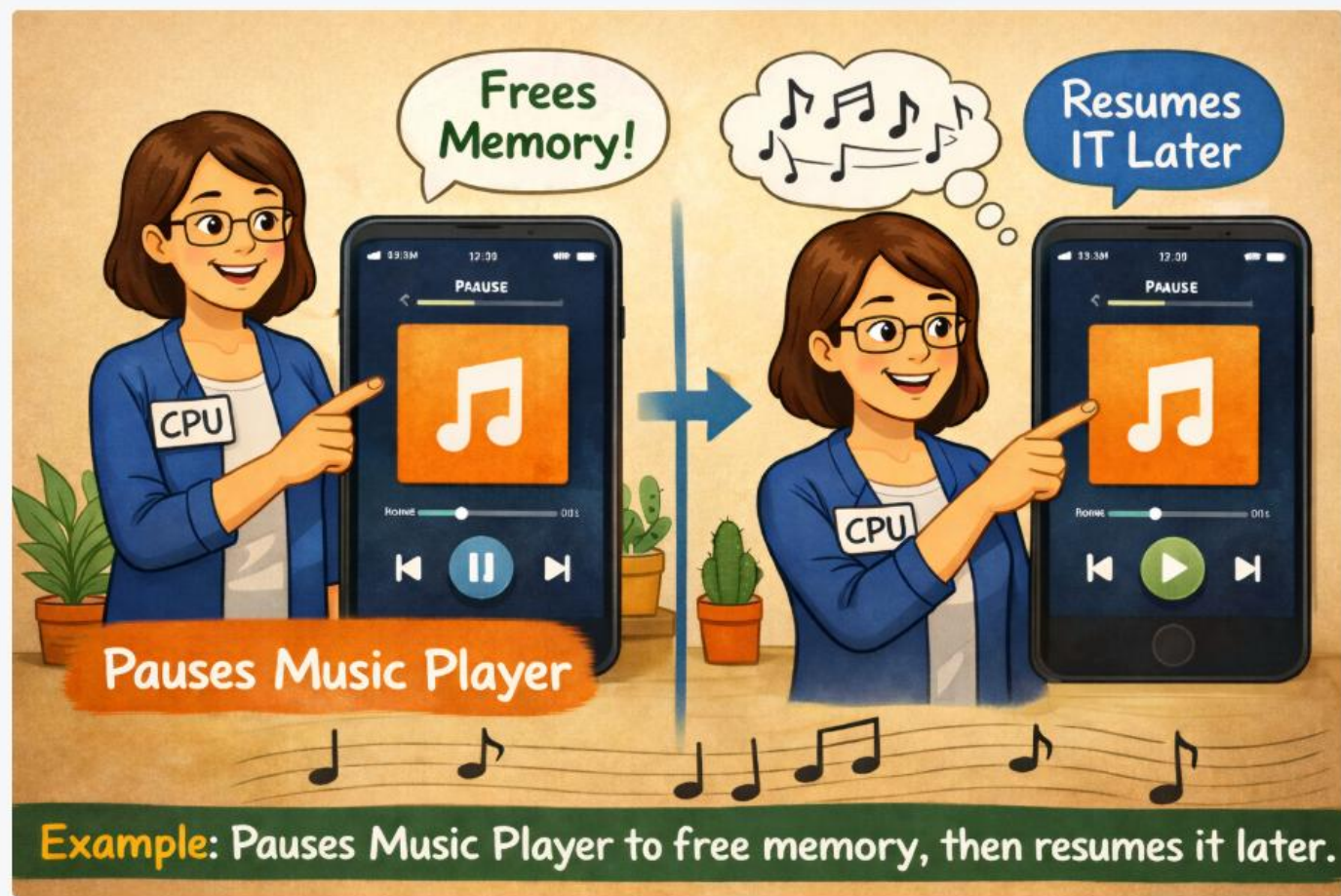
2. Medium-Term Scheduler

What it does:

- **Temporarily removes (suspends)** processes from memory.
- Brings them back later when needed.
- **Example:** Pauses Music Player to free memory, then restarts/resumes it later.



Types of process schedulers



Types of process schedulers

3. Short-Term Scheduler (CPU Scheduler)

- Decides **which ready process gets the CPU next.**
- **Example:** Chooses whether **Browser** or **Word** runs now.

In summary: **Long-term scheduler selects jobs, medium-term scheduler suspends processes, and short-term scheduler assigns the CPU.**

Why do we need to schedule processes?

- The CPU can execute only one process at a time.
- **Several programs are ready to run at the same time, but the CPU can run only one process at any given moment.**
- Scheduling decides which process runs next.

Why do we need to schedule processes?



Why do we need to schedule processes?

Example: You open:

- Web browser
- Word document
- Music player
- All want to run now? But the CPU says:
- “I can run only one process at a time”
- Then, the **scheduler decides** who runs first.

Why do we need to schedule processes?



Objectives of Process Scheduling Algorithm?

- Maximize CPU Utilization
- Minimize Waiting Time
- Reduce how long a process waits in the ready queue.
- **Minimize Turnaround Time**
- Finish processes **as quickly as possible.**
- **Priority Handling**
- Important processes should run **before less important ones.**

Times for Process and CPU Scheduling

1. Arrival Time: Time at which the process arrives in the ready queue.

➤ Example: Process arrives at time **2 ms**.

2. Burst Time (Service Time) : Time required by a process for CPU execution.

➤ Example: Process needs **5 ms** of CPU time.

Times for Process and CPU Scheduling

3. Completion Time (CT)/Finished Time

- Time when a process **finishes execution completely.**

➤ Example: Process finishes at **12 ms**

4. Turnaround Time (TAT)

- Total time spent by a process in the system.

Times for Process and CPU Scheduling

4. Turnaround Time (TAT)

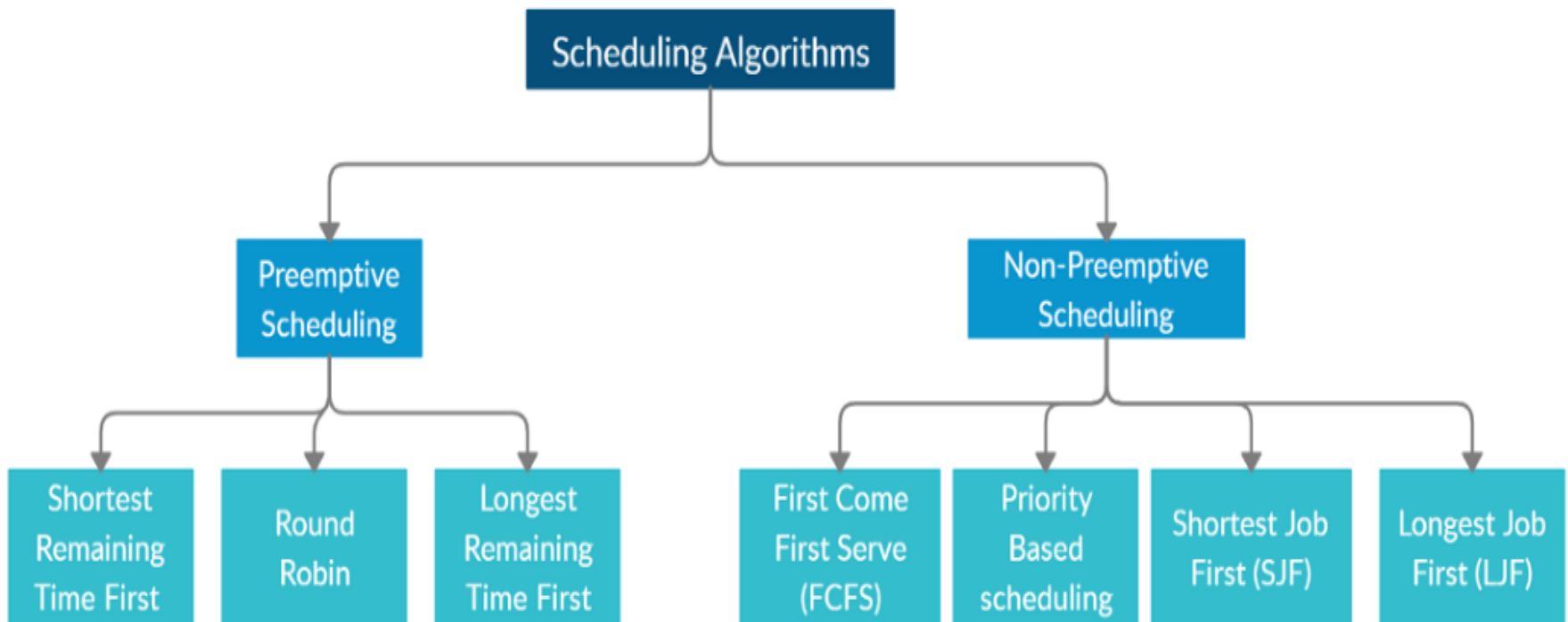
- **Formula:**
- Turnaround Time = Completion Time – Arrival Time.
- Example:
 $CT = 12; AT = 2; \text{ then } TAT = \mathbf{10\ ms}$

Times for Process and CPU Scheduling

5. Waiting Time (WT)

- Time a process spends **waiting in the ready queue.**
- **Formula:**
- $\text{Waiting Time} = \text{Turnaround Time} - \text{Burst time (BT)} / \text{Service Time}.$
- **Example:**
 $\text{TAT} = 10; \text{BT} = 5 ; \text{then } \text{WT} = 5 \text{ ms}.$

Types of CPU Scheduling Algorithms



Types of CPU Scheduling Algorithms

- CPU scheduling can be divided into **two types** based on whether a running process can be **stopped or not**.

1. Preemptive Scheduling:

- The OS can stop (interrupt) a running process.
- CPU is given to another process.

Types of CPU Scheduling Algorithms

When it happens:

- Time quantum expires:
 - Fixed amount of CPU time assigned to a process.
 - Each process gets CPU for **this maximum duration**.
 - If the process **does not finish in this time**, it is preempted (paused) and moved to the ready queue.

Types of CPU Scheduling Algorithms

When it happens:

- **A higher-priority process arrives.**

Real-life example:

- Teacher stops one student mid-sentence to allow another to speak.



Types of CPU Scheduling Algorithms

2. Non-Preemptive Scheduling

- **Once a process gets the CPU, it runs until it finishes or waits for I/O.**
- **OS cannot interrupt it.**
- **Real-life example:** Student speaks until finished, no interruption allowed.



Types of CPU Scheduling Algorithms

- **In summary, Preemptive scheduling allows the OS to interrupt a running process, while non-preemptive scheduling does not.**



Types of CPU Scheduling Algorithms

Preemptive vs Non-Preemptive Scheduling

Preemptive Scheduling	Non-Preemptive Scheduling
The process can be stopped, even before it finishes.	The process is not stopped until it completes its work.
The CPU is given to processes for a set amount of time based on their priority.	CPU is given to the process until it ends.
Tasks are switched based on priority.	No switching happens.
Has extra work involved when switching between processes.	No extra work from switching.
Flexible and adjustable.	Fixed and cannot be changed.
Examples: Shortest Remaining Time First, Round Robin, etc.	Examples: First Come First Serve, Shortest Job First, Priority Scheduling, etc.

Types of CPU Scheduling Algorithms

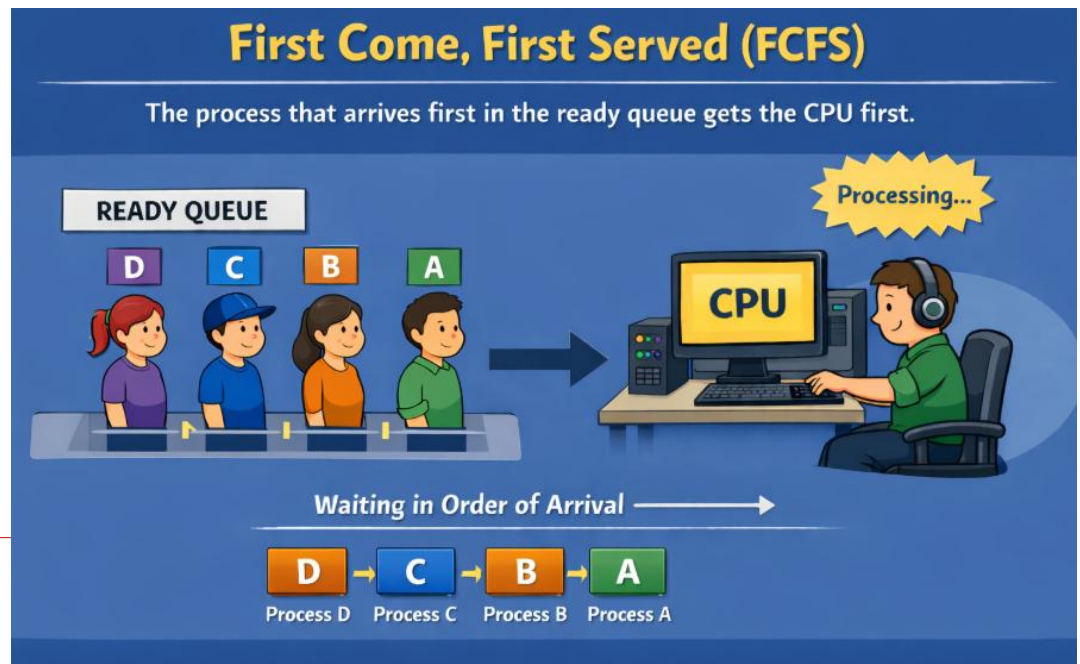
Easy Way to Remember

- **Preemptive = Interrupts allowed, fast, many switches.**
- **Non-preemptive = No interrupts, slow, simple.**



First Come First Serve (FCFS)

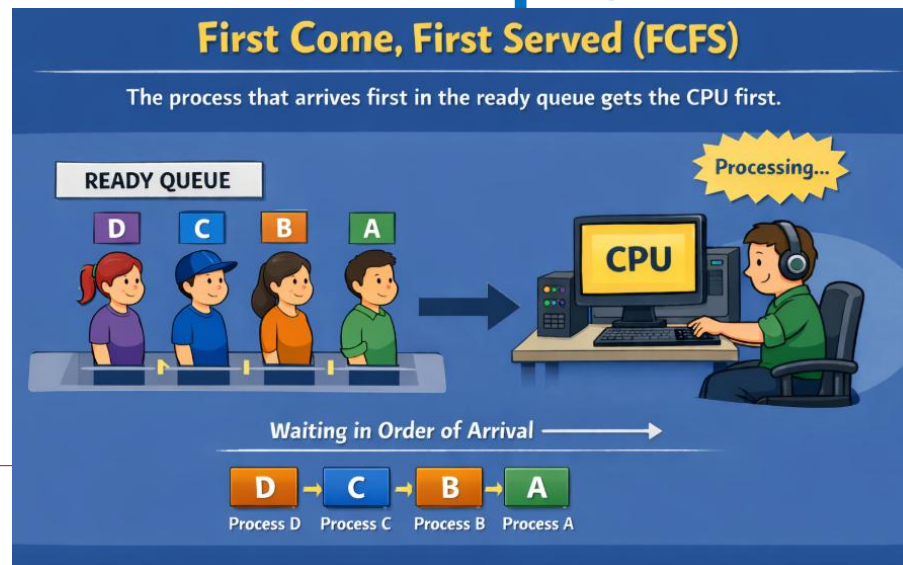
- **The process that arrives first in the ready queue gets the CPU first.**
- Others wait in order of arrival (like a queue at a ticket counter).



First Come First Serve (FCFS)

How it Works (Step by Step):

1. Processes enter the ready queue.
2. CPU picks the **first process in the queue**.
3. Process runs until **completion**.
4. CPU moves to the **next process in order**.



First Come First Serve (FCFS)

- In the **First-Come, First-Served (FCFS)** scheduling algorithm, these terms are defined as follows:
 1. **AT (Arrival Time):** The time at which a process enters the ready queue.
 2. **BT (Burst Time)/Service Time :** The total time required by a process to complete execution on the CPU.
 3. **CT (Completion Time):** The time at which a process finishes its execution.

First Come First Serve (FCFS)

4. WT (Waiting Time): The time a process spends waiting in the queue before getting the CPU.

5. TAT (Turnaround Time): The total time taken from the arrival to the completion of a process.

6. (Response Time): is the duration from when a process enters the ready queue until it gets the CPU for the first time.

Formula: $TAT = CT - AT$; $RT=WT$.

Example 1

First Come First Serve (FCFS)

Process	Arrival Time	Burst Time	Completion Time	Turnaround Time (TAT)	Waiting Time (WT)	Response Time (RT)
P4	0	6	6	$6 - 0 = 6$	$6 - 6 = 0$	0
P2	1	5	11	$11 - 1 = 10$	$10 - 5 = 5$	5
P1	2	1	12	$12 - 2 = 10$	$10 - 1 = 9$	9
P5	2	3	15	$15 - 2 = 13$	$13 - 3 = 10$	10
P3	4	1	16	$16 - 4 = 12$	$12 - 1 = 11$	11

Completion Time (CT) = Arrival Time + Burst Time/Service Time.

Turnaround Time (TAT) = CT – AT.

Waiting Time (WT) = TAT – BT.

RT=WT.

Example 2

Process	Arrival Time	Burst Time	Completion Time (CT)	Turnaround Time (TAT)	Waiting Time (WT)	Response Time (RT)
P1	0	4	4	$4 - 0 = 4$	$4 - 4 = 0$	0
P2	1	3	7	$7 - 1 = 6$	$6 - 3 = 3$	3
P3	2	2	9	$9 - 2 = 7$	$7 - 2 = 5$	5
P4	3	1	10	$10 - 3 = 7$	$7 - 1 = 6$	6

Completion Time (CT) = Arrival Time + Burst Time/Service Time.

Turnaround Time (TAT) = CT - AT.

Waiting Time (WT) = Turnaround Time (TAT) - Burst Time (BT).
RT = WT.

Shortest Job Next (SJN)

- The process **Shortest Job Next (SJN)** is a CPU scheduling algorithm where the **process with the smallest burst time is executed first.**
- **Key Idea :** Do the shortest work first.

Shortest Job Next (SJN)

How SJN Works

- All processes enter the ready queue.
- CPU selects the process with the **shortest burst time**.
- Process runs until it finishes.
- Repeat for remaining processes.

Shortest Job Next (SJN)

■ Example Process Table

Process	Arrival Time (AT)	Burst Time (BT)
P1	0	5
P2	1	3
P3	2	2
P4	3	4

Execution Order (SJN):

- At time 0 → only **P1** is available → runs first.
After P1 finishes, choose the shortest job among.
remaining: **Order:** P1 → P3 → P2 → P4.

Shortest Job Next (SJN)

Process	Arrival Time	Burst Time	Completion Time (CT)	Turnaround Time (TAT)	Waiting Time (WT)	Response Time (RT)
P1	0	5	5	5	0	0
P3	2	2	7	5	3	3
P2	1	3	10	9	6	6
P4	3	4	14	11	7	7

Completion Time (CT)=Arrival Time + Burst Time/Service Time.

Turnaround Time : $TAT = CT - AT$.

Waiting Time (WT) = Turnaround Time (TAT) – Burst Time (BT)
RT=WT.

LJF (Longest Job First) Scheduling

- **Longest Job First (LJF)** is a CPU scheduling algorithm where the **process with the largest burst time is executed first.**

How LJF Works

- Processes arrive in the ready queue.
- CPU selects the process with the **longest burst time.**
- That process runs until completion.
- Repeat for remaining processes.

LJF (Longest Job First) Scheduling

◆ Example (LJF – Non-Preemptive)

Given Process Table

Process	Arrival Time	Burst Time
P1	0	4
P2	1	6
P3	2	2
P4	3	5

- **Gantt Chart (LJF)**
- At time 0, only **P1** is available → runs first
After that, choose the **longest job** among remaining. **Order:** P1 → P2 → P4 → P3.

	P1		P2		P4		P3	
0		4		10		15		17

LJF (Longest Job First) Scheduling

Process	Arrival Time	Burst Time	Completion Time (CT)	Turnaround Time (TAT)	Waiting Time (WT)	Response Time (RT)
P1	0	4	4	$4 - 0 = 4$	$4 - 4 = 0$	0
P2	1	6	10	$10 - 1 = 9$	$9 - 6 = 3$	3
P4	3	5	15	$15 - 3 = 12$	$12 - 5 = 7$	7
P3	2	2	17	$17 - 2 = 15$	$15 - 2 = 13$	13

Completion Time (CT)=Arrival Time + Burst Time/Service Time.

Turnaround Time : $TAT = CT - AT$.

Waiting Time (WT) = Turnaround Time (TAT) – Burst Time (BT)
RT=WT.

Practice time

- Python Programming Language based on OS